

LAB 2: MICROSOFT WINDOWS ADMINISTRATION

CNIT24200-03

Group 3

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PROCEDURES

WoodRock was a wood sales company. Active Directory (AD) parent domain group03.c24200.cit.lcl focused on the Accounting department, Human Resources department, and Executive department organization. AD sub domain c242-03-b.group03.c24200.cit.lcl focused on the Warehouse department, Shipping department, and Marketing department organization. AD sub domain c242-03-b.group03.c24200.cit.lcl focused on the Outside Sales department, Inside Sales department, and Account Management Department organization. Computer G03SRV01 was responsible for the AD parent domain group03.c24200.cit.lcl administration. Computer G03SRV02 was responsible for the AD domain c242-03-b.group03.c24200.cit.lcl administration. Computer G03SRV03 was responsible for the AD domain c242-03-a.group03.c24200.cit.lcl administration. The following tasks assigned to each member of group 03 were Organizational Units (OUs) creation, OU's Domain Local and Global Groups creation, Users creation, Password Policy, Partition/Volume creation, Shared Resources Right Grant, Network File Sharing for Shared Resources, VMware Workstation installation, Windows 11 Virtual Machine (VM) installation, Windows 11 VM configuration, Group Policy Object (GPO) Setup for "regedit.msc" disablement, Nesting Global Groups in Read & Modifying Groups, Windows 11 VM Domain Name System (DNS) & Internet Protocol (IP) configuration, Windows 11 VM promotion, Business Case GPO, User Disk Quotas Setup, Domain Local (DL) User Security Group Creation, Folder Redirection & Roaming Profiles Creation, Folder Redirection GPO, Roaming Profiles GPO, Network Printing implementation, Printer GPO, Configuration testing, and Windows Server Activation. Each section contained a numbered list which showed the steps. Sections' headers, which were bolded and centered, showed the actions taken. The bibliography appeared at the end of the report, followed by Appendix A, which contained the problem statement. Additional supporting documentation was included

in subsequent appendices. The followings were the formatting keys: **buttons** were bold, *options* were italicized, text entered into the computer was in Courier New, menu navigation was by the pipe symbol and italic words: *Start | Programs | MS Office | Word*. Terms enclosed in square brackets indicated definitions with multiple interpretations: [workstation's name] referred to G03SRV01, G03SRV02, or G03SRV03. Each [definition] was provided once in the introduction of the relevant section and was not repeated.

Organizational Units Creation

Organizational Units (OUs) were mainly used in business to organize objects (mainly users and computers) in Active Directory (AD); to better understand their implementation, Organizational Units Creation demonstrated how the Organizational Units were created on each Active Directory (AD) domain. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. [AD domain] were interpreted as group03.c24200.cit.lcl, c242-03-b.group03.c24200.cit.lcl, or c242-03-a.group03.c24200.cit.lcl which accorded to the domain name on the computer the user was using. [department name] were interpreted as the name of the department which accorded to the user's business scenario. There are 3 departments in each AD domain. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Active Directory Users and Computers*
4. Right-clicked *[AD domain]*
5. Clicked *New | Organizational Unit*
6. Typed `[department name]`
7. Clicked *OK* in *New Object - Organizational Unit* window
8. Repeated steps 4-7 for each `[department name]`
9. Repeated steps 1-8 for G03SRV01 and G03SRV02

Organizational Unit's Security Groups Creation

Security Groups were used for better permissions organizations. Group type, Domain Local (DL), was intended to be used only within the domain where the group was created; but, the group could contain universal groups from any domain, and domain local groups from the same domain. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. [OU] was interpreted as the name of the OU created which was uniquely assigned to [department name]. Inside each OU, there are 3 security groups. One security group, named, [department name]_READ_DL, was assigned with the read permission to the shared resources (shown in the later section) and one security group, named [department name]_MOD_DL, was assigned for the write permission to the shared resources. One security group, named [department name]_MANAGER_G, was nested in [department name]_READ_DL and [department name]_MOD_DL to inherit read and write permissions, respectively. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Active Directory Users and Computers*
4. Right-clicked *[OU] | New | Group*
5. Selected *Domain local | Security*
6. Typed `[department name]_READ_DL` in *Group name*
7. Clicked *OK* in *New Object - Group* window
8. Repeated steps 4-7; however, typed `[department name]_MOD_DL` in *Group name* at step 6

9. Repeated steps 4-7; however, typed [department name]_MANAGER_G in *Group name* at step 6 and selected *Global* | *Security* at step 5
10. Right-clicked [department name]_MANAGER_G
11. Clicked *Properties* | *Member Of* | *Add...* | *Locations...* | *[AD Domain]* | *OK* in *Locations* window | *Advanced...* | *Find Now*
12. Selected *[department name]_READ_DL* and *[department name]_MOD_DL* (held **Ctrl** to select multiple options)
13. Clicked *OK* in *Select Groups* window | *OK* in *Select Groups* window | *OK* in *[department name]_MANAGER_G Properties* window
14. Repeated steps 4-14 for each [OU]
15. Repeated steps 1-15 for G03SRV01 and G03SRV02

Users Creation

Users were created within a specific domain and can authenticate against any Domain Controller (DC) within the domain; security groups were container objects. Security groups were assigned to certain rights, which could delegate the rights to the leaf objects contained. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. There were a total of 6 security groups in each AD domain. [security group] was interpreted as the name of the security group, which could be [department name]_READ_DL, [department name]_MOD_DL, or [department name]_MANAGER_G. Each security group created in the above sections were expected to have at least 2 users, which led to the total of 18 user objects created in each AD domain. Subsequently, every 2 newly-created users were expected to belong to the same security group. [newly-created user object] refers to the most recently created user object. [user's first name] was interpreted as the first name of the user assigned to the user object; likewise, [user's initials], [user's last name], [user's logon name], [user's password] were interpreted as user's middle initial, user's last name, user's logon name, and user's password, respectively. [user's logon name] was expected to not be repeated. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Active Directory Users and Computers*
4. Right-clicked *[OU]*
5. Clicked *New | User*
6. Typed [user's first name] in *First name*, [user's initials] in *Initials*, [user's last name] in *Last name*, [user's logon name] in *User logon name*

7. Typed [user's password] in *Password* and in *Confirm password*
8. Clicked *Next* | *Finish*
9. Repeated steps 4-8 for a different [user's logon name]
10. Right-clicked [*security group*]
11. Clicked *Properties* | *Member* | *Add...* | *Locations...* | [*AD Domain*] | [*OU*] | *OK* in *Locations* window | *Advanced...* | *Find Now*
12. Selected 2 [*newly-created user object*] (held **Shift** to select multiple options)
13. Clicked *OK* in *Select Groups* window | *OK* in *Select Groups* window | *OK* in [*security group*] *Properties* window
14. Repeated steps 4-13 for each [*security group*] in [*OU*]
15. Repeated steps 4-14 for each [*OU*]
16. Repeated steps 1-15 for G03SRV01 and G03SRV02

Password Policy

Passwords could be a powerful method of authentication if properly managed. Generally, password policy was defined through a GPO. The goal of the password policy section was to demonstrate password policy setup through GPO. Adjustment could be made accordingly. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion.

Windows Server 2022 was expected to be booted before the first step was started.

1. Opened *Server Manager*
2. Clicked *Tools | Group Policy Management | [AD domain]*
3. Right-clicked *Default Domain Policy | Edit...*
4. Clicked *Computer Configuration | Policies | Windows Settings | Security Settings | Password Policy*
5. Right-clicked *Minimum password length | Properties*
6. Typed 8 in *characters*
7. Clicked *OK*
8. Closed *Group Policy Management Editor*
9. Right-clicked *Default Domain Policy | Enforced*
10. Closed *Group Policy Management*
11. Repeated steps 1-11 for G03SRV01 and G03SRV02

Shared Resources Right Grant & Partition Creation

Each OU had its own way of sharing resources. At a basic level, one security group had read access to the folder and another security group had write access to the folder. Every user object or group object inside the designated security which was assigned with read permission had read permission; likewise, every user object or group object inside the designated security which was assigned with write permission had read permission.

Commonly, security groups which needed both read and write permissions could be nested in the previously-mentioned 2 security groups. Adjustment could be made accordingly. The Shared Resources Right Grant & Partition Creation section demonstrated shared resources right grant. [folder] referred to the folder created which was uniquely assigned to [OU] in [AD domain]. Each folder was configured with New Technology File System (NTFS) permissions. [unnecessary users] referred to users which were usually not permitted in permissions entries or Access Control Entry (ACE), which were neither Administrators nor SYSTEM. The folders were created in the 1-terabyte (TB) disk. The Shared Resources Right Grant & Partition Creation section also demonstrated how a new partition/volume was created. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Right-clicked *Window* icon in left bottom of the screen | *Disk Management* | *Unallocated space* under the black bar | *New Simple Volume...*
2. Clicked *Next >* in *Welcome to the New Simple Volume Wizard* window | *Next >* in *Specify Volume Size* window | *Next >* in *Assign Drive Letter or Path* window | *Next >* in *Format Partition* window | *Finish*
3. Clicked *File Explorer* in *Windows Taskbar* | *New Volume (D:)*
4. Pressed **Ctrl + Shift + N**

5. Typed [department name]
6. Right-clicked [folder] | *Properties*
7. Clicked *Security* | *Advanced* | *Disable inheritance* | *Convert inherited permissions into explicit permissions on this object*
8. Clicked [unnecessary users] | *Remove*
9. Repeated step 9 until all [unnecessary users] were removed
10. Clicked *Add* | *Select a principal* | *Locations...* | [AD Domain] | [OU] | *OK* in *Locations* window | *Advanced...* | *Find Now*
11. Selected [department name]_READ_DL
12. Clicked *OK* in *Select User, Computer, Service Account, or Group* window | *OK* in *Select User, Computer, Service Account, or Group* window | *OK* in *Select User, Computer, Service Account, or Group* window | *Write* | *OK* in *Permission Entry for [folder]*
13. Repeated steps 10-12 for [department name]_MOD_DL
14. Clicked *OK* in *Advanced Security Settings for [folder]* | *OK* in [folder] *Properties*
15. Repeated steps 4-14 for each [OU]
16. Repeated steps 1-15 for G03SRV01 and G03SRV02

VMware Workstation Installation

The section demonstrated how VMware Workstation installation was done on the workstation; subsequently, the VMware Workstation was used in the later section, which demonstrated how Windows 11 VM was installed. The VMware Workstation was also used for testing purposes, which was demonstrated at the ending section. Likewise, the VMware Workstation Installation section also demonstrated how the rtfm network drive was mapped. [school account username] and [school account password] referred to the username and password of the school account which was authorized through the Computer Information and Technology (CIT) domain, which was necessary for rtfm network drive access. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows + E**
2. Right-clicked *This PC | Map network drive*
3. Typed `\\rtfm.cit.lcl`
4. Clicked *Browse... | rtfm.cit.lcl*
5. Typed `CIT\[school account username]` in *User Name*
6. Typed `[school account password]` in *Password*
7. Clicked *rtfm.cit.lcl | ISO | Windows | Client | Windows 11 Old | OK* in *Browse For Folder* window | *Finish* |
en-us_windows_11_consumer_editions_version_24h2_x64_dvd_1d5fcad3
8. Pressed **Ctrl + C**
9. Clicked *New Volume (D:)*
10. Pressed **Ctrl + V**
11. Repeated steps 3-4

12. Clicked *rtfm.cit.lcl* | *ISO* | *Windows* | *Pub* | *CNIT24200* | *OK* in *Browse For Folder* window | *Finish* | *VMware-workstation-full-17.6.2-24409262*
13. Repeated steps 8-10
14. Right-clicked *VMware-workstation-full-17.6.2-24409262* | *Open*
15. Clicked *Next* in *Welcome to the VMware Workstation Pro Setup* window | *I accept the terms in the License Agreement* | *Next* in *Compatible Setup* window | *Next* in *Custom Setup* | *Next* in *User Experience Settings* | *Next* in *Shortcuts* | *Install*
16. Repeated steps 1-15 for G03SRV01 and G03SRV02

Windows 11 Virtual Machine Installation

The section demonstrated how Windows 11 VM Installation was done on the VMware Workstation; hence, VMware Workstation was required to be preinstalled. Windows 11 VM was also used for testing purposes, which was demonstrated at the ending section. [VM] referred to the name of the VM created; hence, [VM] could be G03S1WIN11, G03S2WIN11, or G03S3WIN11, which was uniquely assigned to the workstation the user was on, G03SRV01, G03SRV02, or G03SRV03, respectively. [VM password] referred to the password for VM encryption. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started. New Volume (D:) was expected not to have any VMs prior to Windows 11 VM Installation.

1. Pressed **Windows**
2. Typed `VMware Workstation Pro`
3. Clicked *VMware Workstation Pro | Create a New Virtual Machine | Next >* in *Welcome to the New Virtual Machine Wizard* window | *Installer disc image file (iso) | Browse... | New Volume (D:) | Open | Next >* in *Guest Operating System Installation* window
4. Typed [VM]
5. Clicked *Browse... | New Volume (D:) | OK* in *Browse For Folder* window | *Next >* in *Name the Virtual Machine* window
6. Typed [VM password] in *Password and Confirm Password*
7. Clicked *Next >* in *Encryption Information* window | *Split virtual disk into multiple files | Next >* in *Specify Disk Capacity* window | *Finish*
8. Pressed **Enter** in [VM] window | **Enter**

9. Closed *[VM]* - *VMware Workstation* window
10. Clicked *Power Off* in *VMware Workstation* window
11. Repeated steps 1-9 for G03SRV01 and G03SRV02

Windows 11 VM Configuration

The section demonstrated how Windows 11 VM Configuration was done on the Windows 11 VM; hence, Windows 11 VM Installation was expected to be done prior to Windows 11 VM Configuration. Windows 11 VM was also used for testing purposes, which was demonstrated at the ending section. [VM configuration name] and [VM configuration password] referred to the name and password of the user during the VM configuration process, respectively. [security question option] referred to the option to the security question, which could be “What’s the name of the city where you were born?”, “What’s the name of the city where your parents met?”, “What’s the first name of your oldest cousin?”, and “What’s the name of the first school you attended?”. [security question number] referred to the order number of the security questions the user had to answer, which could be 1, 2, or 3. [security question answer] referred to the answer to the [security question option] which the user picked. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion.

Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `VMware Workstation Pro`
3. Clicked *VMware Workstation Pro | Power on this virtual machine | Next in Select language settings window | Next in Select keyboard settings window | I agree everything will be deleted including files, apps, and settings | Next in Select setup option*
4. Typed `NW6C2-QMPVW-D7KKK-3GKT6-VCFB2` in *Enter Product key*
5. Clicked *Next in Product key window | Accept | Next in Select location to install Windows 11 | Install | Yes in Is this the right country or region? window | Yes in Is this*

the right keyboard layout or input method? window | *Skip* in *Want to add a second keyboard layout?* window | *Sign-in options* | *Domain join instead*

6. Typed *[VM configuration name]* in *Enter your name*
7. Clicked *Next* in *Who's going to use this device* window
8. Typed *[VM configuration password]* in *Enter a password*
9. Clicked *Next* in *Create a super memorable password* window
10. Typed *[VM configuration password]* in *Password confirmation*
11. Clicked *Next* in *Confirm your password* window
12. Typed *[security question answer]* in *Your answer*
13. Clicked *Security question ([security question number] of 3)* | *[security question option]* | *Next* in *Now add security questions* window
14. Repeated step 13 for each *[security question number]*
15. Clicked *Next* in *Choose privacy settings for your device* window | *Next* in *Choose privacy settings for your device* window | *Next* in *Choose privacy settings for your device* window | *Accept*
16. Repeated steps 1-15 for G03SRV01 and G03SRV02

Windows 11 VM DNS Configuration

The section demonstrated how Windows 11 VM DNS Configuration was done on the Windows 11 VM; hence, Windows 11 VM Configuration was expected to be done prior to the Windows 11 VM DNS Configuration section; likewise, VMware Workstation was expected to be opened following Windows 11 VM Configuration. [Domain Controller IP address] could be interpreted as 44.38.3.2, 44.38.3.3, or 44.38.3.4, which was uniquely assigned to the computer the user was on, G03SRV01, G03SRV02, or G03SRV03, respectively. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion.

Windows Server 2022 was expected to be booted before the first step was started.

1. Clicked *VMware Workstation Pro* in *Windows Taskbar*
2. Clicked *[VM]* screen
3. Typed *[VM configuration password]*
4. Pressed **Enter**
5. Pressed **Windows**
6. Typed *View Network Connections*
7. Clicked *View Network Connections*
8. Right-clicked *Ethernet0*
9. Clicked *Properties | Internet Protocol Version 4 (TCP/IPv4) | Properties | Use the following DNS server addresses*
10. Typed *[Domain Controller IP address]* in *Preferred DNS server*
11. Clicked *OK* in *Internet Protocol Version 4 (TCP/IPv4) Properties* window | *Close*
12. Closed *Network Connections*
13. Repeated steps 1-12 for G03SRV01 and G03SRV02

Windows 11 VM Computer Object

The section demonstrated how Windows 11 VM was promoted into the AD domain the user was on. As part of the Windows 11 VM Computer Object section, the computer's name of the VM was changed. [administrator password] referred to the password which was uniquely assigned to the administrator user inside the [AD domain]. After the Windows 11 VM Computer Object section was done, Windows 11 VM could be viewed as a computer object inside the server manager. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows 11 VM DNS & IP Configuration was expected to be done prior to the Windows 11 VM Computer Object section; likewise, Windows Server 2022 was expected to be opened, VM user was expected to be logged in, and VMware Workstation was expected to be opened following before the first step was started.

1. Clicked *VMware Workstation Pro* in *Windows Taskbar*
2. Clicked *[VM]* screen
3. Pressed **Windows + R**
4. Typed `sysdm.cpl`
5. Pressed **Enter**
6. Clicked *Computer Name | Change... | Domain*
7. Typed *[VM]* in *Computer name* and *[AD domain]* in *Domain*
8. Clicked *OK* in *Computer Name/Domain Changes*
9. Typed `Administrator` in *User name*
10. Typed `[administrator password]` in *Password*
11. Clicked *OK* in *Computer Name/Domain Changes* window | *OK* in *Computer Name/Domain Changes* window | *Close* | *Restart Now*
12. Closed *[VM]* - *VMware Workstation* window

13. Repeated steps 1-12 for G03SRV01 and G03SRV02

Windows 11 VM IP Address Configuration

The section demonstrated how the Windows 11 VM IP address was configured on the computer the user was on. As part of the Windows 11 VM IP Address Configuration section, the computer's IP address of the VM was changed. [VM IP address] referred to the IP address the VM was supposed to be configured. [VM IP address] could be interpreted as 44.38.3.5, 44.38.3.6, or 44.38.3.7, which was uniquely assigned to the VM in the computer the user was on, G03SRV01, G03SRV02, or G03SRV03, respectively. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows 11 VM Computer Object was expected to be done prior to the Windows 11 VM IP Address Configuration section.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Active Directory Users and Computers | Computers*
4. Right-clicked *[VM]*
5. Clicked *Properties | Dial-in | Assign Static IP Address | Assigned a static IPv4 address*
6. Typed `[VM IP address]`
7. Clicked *OK* in *Static IP Address* | *OK* in *[VM] Properties*
8. Repeated steps 1-7 for G03SRV01 and G03SRV02

Registry Editor Disabling

The section demonstrated how Registry Editor Disabling was done on Windows Server 2022 with GPO setup. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Group Policy Management*
4. Right-clicked *[AD domain] | Create a GPO in this domain, and Link it here...*
5. Typed `Disable Regedit` in *Name*
6. Clicked *OK* in *New GPO* window
7. Right-clicked *Disable Regedit | Edit...*
8. Expanded *User Configuration | Policies | Administrative Template | System*
9. Right-clicked *Prevent access to registry editing tools*
10. Clicked *Edit | Enabled | Apply | OK*
11. Right-clicked *Disable Regedit | Enforced*
12. Repeated steps 1-10 for G03SRV01 and G03SRV02

Business Case GPO

The section demonstrated how Business Case GPO was created and configured on Windows Server 2022. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Group Policy Management*
4. Right-clicked *[AD domain] | Create a GPO in this domain, and Link it here...*
5. Typed `Business Case in Name`
6. Clicked *OK* in *New GPO* window
7. Right-clicked *Business Case | Edit...*
8. Expanded *Computer Configuration | Policies | Windows Settings | Security Settings | Local Policies | Security Options*
9. Right-clicked *Interactive logon: Machine inactivity limit*
10. Clicked *Properties | Defined this policy setting*
11. Typed `600` in *seconds*
12. Clicked *OK* in *Interactive logon: Machine inactivity limit Properties* window
13. Closed *Group Policy Management Editor* window
14. Right-clicked *Business Case | Enforced*
15. Repeated steps 1-14 for G03SRV01 and G03SRV02

Redirected Folders GPO

The section demonstrated how Redirected Folders GPO was created and configured on Windows Server 2022; however, Redirected Folders GPO setup alone was not sufficient because the Universal Naming Convention which was configured as a redirected folder path was not a valid path. However, section Network File Sharing for Shared Resources and section NTFS Permissions Setup for RedirectedFolders & Profiles\$ resolved the issue; thus, the Redirected Folders GPO section was required to be completed prior to both sections. [workstation] referred to the name of the computer the user was on. [workstation] can be interpreted as G03SRV03, G03SRV02, or G03SRV01. [redirected folder] referred to the type of user's folder which was redirected, which could be Desktop or Documents. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Group Policy Management*
4. Right-clicked *[AD domain] | Create a GPO in this domain, and Link it here...*
5. Typed `Redirected Folders` in *Name*
6. Clicked *OK* in *New GPO* window
7. Right-clicked *Redirected Folders | Edit...*
8. Expanded *User Configuration | Policies | Windows Settings | Folder Redirection*
9. Right-clicked *[redirected folder]*
10. Clicked *Properties | Not configured | Basic - Redirect everyone's folder to the same location*
11. Typed `\\[workstation]\RedirectedFolders` in *Root Path*

12. Clicked *OK* in *[redirected folder] Properties* window | *Yes* in *Warning* window |
Properties | *Edit* | *Enabled* | *Apply* | *OK*
13. Repeated steps 9-12 for another [redirected folder]
14. Closed *Group Policy Management Editor* window
15. Right-clicked *Redirected Folders* | *Enforced*
16. Repeated steps 1-15 for G03SRV01 and G03SRV02

Roaming Profiles GPO

The section demonstrated how Roaming Profiles GPO was created and configured on Windows Server 2022; however, Roaming Profiles GPO setup alone was not sufficient because the Universal Naming Convention which was configured as a roaming profile path was not a valid path. However, section Network File Sharing for Shared Resources and section NTFS Permissions Setup for RedirectedFolders & Profiles\$ resolved the issue; thus, the Roaming Profiles GPO section was required to be completed prior to both sections. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Group Policy Management*
4. Right-clicked *[AD domain] | Create a GPO in this domain, and Link it here...*
5. Typed `Roaming Profiles` in *Name*
6. Clicked *OK* in *New GPO* window
7. Right-clicked *Roaming Profiles | Edit...*
8. Expanded *Computer Configuration | Policies | Administrative Template | System | User Profiles*
9. Right-clicked *Set roaming profile path for all users logging onto this computer*
10. Clicked *Edit | Enabled*
11. Typed `\\[workstation]\Profiles$\%USERNAME%` in *Users logging onto this computer should use this roaming profile path*
12. Clicked *OK* in *Set roaming profile path for all users logging onto this computer* window

13. Closed *Group Policy Management Editor* window
14. Right-clicked *Roaming Profiles* | *Enforced*
15. Repeated steps 1-14 for G03SRV01 and G03SRV02

User Disk Quotas Setup

User Disk Quotas were tools which monitored, managed, and limited the amount of disk space a user could access on a file system. The User Disk Quotas Setup section demonstrated how folders created from redirected folders and roaming profiles were created and how quota policies were enforced on said folders. Disk space was denied to users exceeding 1-gigabyte and warnings were given to users surpassing 900mbs. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows + E**
2. Clicked *This PC | New Volume (D:)*
3. Pressed **Ctrl + Shift + N**
4. Typed name `RedirectedFolders`
5. Right-clicked *New Volume (D:)*
6. Repeated step 3
7. Typed `Profiles$`
8. Right-clicked *New Volume (D:)*
9. Clicked *Properties | Quota*
10. Checked *Enable Quota Management | Deny disk space to users exceeding quota limit | Log event when a user exceeds their quota limit | Log event when a user exceeds their warning level*
11. Clicked *Limit disk space to*
12. Type 1 in *Limit disk space to* and 900 in *Set warning level to*
13. Clicked *OK* in *New Volume (D:) Properties*
14. Repeated steps 1-13 for G03SRV01 and G03SRV02

Network File Sharing for Shared Resources

Section Shared Resources Right Grant & Partition Creation demonstrated how each OU had its own way of sharing resources and how NTFS permissions were configured for each folder; subsequently, the Network File Sharing for Shared Resources section demonstrated how each [folder] created and assigned uniquely for each [OU] was enabled for network file sharing and configured for share permissions; likewise, RedirectedFolders and Profiles\$ were done the same way as [folder], which was shown in the Network File Sharing for Shared Resources section as well. Section Disk Quota Setup was expected to be done prior to the Network File Sharing for Shared Resources section. Adjustment could be made accordingly. [network file sharing folder] referred to the folder which was configured to be shared across [AD domain]. [network file sharing folder] could be [folder], RedirectedFolders, or Profiles\$. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Clicked *File Explorer* in *Windows Taskbar* | *New Volume (D:)*
2. Right-clicked [network file sharing folder] | *Properties*
3. Clicked *Sharing* | *Advanced Sharing...* | *Share this folder* | *Permissions* | *Remove* | *Add...* | *Locations...* | *Advanced...* | *Find Now*
4. Held **Ctrl**
5. Selected [department name]_READ_DL and [department name]_MOD_DL
6. Released **Ctrl**
7. Repeated steps 4-6 for each [department name]
8. Clicked *OK* in *Select User, Computer, Service Account, or Group* window | *OK* in *Select User, Computer, Service Account, or Group* window | *OK* in *Select User,*

Computer, Service Account, or Group window | *Change* | *OK* in *Permissions for [network file sharing folder]* | *OK* in *Advanced Sharing* window | *Close*

9. Repeated steps 2-8 for each [network file sharing folder]
10. Repeated steps 1-9 for G03SRV01 and G03SRV02

NTFS Permissions Setup for RedirectedFolders & Profiles\$

Section Shared Resources Right Grant & Partition Creation demonstrated how each OU had its own way of sharing resources and how NTFS permissions were configured for each folder; likewise, RedirectedFolders and Profiles\$ were configured almost the same way as [folder], which was shown in the Shared Resources Right Grant & Partition Creation section. However, unlike [folder], RedirectedFolders and Profiles\$ had broader users, which were all DL users inside [AD domain]. Section Disk Quota Setup was expected to be done prior to the NTFS Permissions Setup for RedirectedFolders & Profiles\$ section. [seamless profile folder] referred to RedirectedFolders or Profiles\$. Adjustment could be made accordingly. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Clicked *File Explorer* in *Windows Taskbar* | *New Volume (D:)*
2. Right-clicked *[seamless profile folder]*
3. Clicked *Properties* | *Security* | *Advanced*
4. Clicked *[unnecessary users]* | *Remove*
5. Repeated step 4 until all [unnecessary users] were removed
6. Clicked *Add* | *Select a principal* | *Locations...* | *Advanced...* | *Find Now*
7. Selected *[department name]_READ_DL*
8. Clicked *OK* in *Select User, Computer, Service Account, or Group* window | *Write* | *OK* in *Permission Entry for [seamless profile folder]*
9. Repeated steps 6-8 for *[department name]_MOD_DL*
10. Repeated steps 6-9 for each [department name]
11. Clicked *OK* in *Advanced Security Settings for [seamless profile folder]* window | *OK* in *[seamless profile folder] Properties*

12. Repeated steps 2-11 for another [seamless profile folder]
13. Repeated steps 1-12 for G03SRV01 and G03SRV02

Network Printer Implementation

Network Printer Implementation showed how the Print Management Tool was installed on Window Server 2022 and how the printer was connected through the Print Management Tool. Adjustment could be made accordingly. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager* | *Manage* | *Add Roles and Features* | *Next >* in *Before you begin* window | *Next >* in *Installation Type* window | *Next >* in *Server Selection* window
4. Checked *Print and Document Services*
5. Clicked *Add Features* | *Next >* in *Server Roles* window | *Next >* in *Features* window | *Next >* in *Print and Document Services* window | *Next >* in *Role Services* window | *Install* | *Close* | *Tools* | *Print Management* | *Print Servers* | *G03SRV03 (local)*
6. Right-clicked *Printers* | *Add Printer...*
7. Clicked *Add an IPP, TCP/IP, or Web Services Printer by IP address or hostname* | *Next >* in *Printer Installation* window
8. Typed `44.3.1.238`
9. Clicked *IPP Printer* | *Autodetect* | *Next >* in *Printer Address* window
10. Typed `k238` in *Printer Name* and *Share Name*
11. Clicked *Next >* in *Printer Name and Sharing Settings* window | *Next >* in *Printer Found* | *Finish*
12. Closed *Print Management* window

13. Repeated steps 1-12 for G03SRV01 and G03SRV02

Printer GPO

Printer GPO showed how the Printer GPO was created and how the printer was shared across [AD domain]. Adjustment could be made accordingly. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Tools | Group Policy Management*
4. Right-clicked *[AD domain] | Create a GPO in this domain, and Link it here...*
5. Typed `printers` in *Name*
6. Clicked *OK* in *New GPO* window
7. Right-clicked *printers | Edit...*
8. Expanded *User Configuration | Preferences | Control Panel Settings | Printers*
9. Clicked *Add a new item | ... | [workstation]-k238 (HP LaserJet E40040) | OK* in *Find Customer Search* window | *Common | Item-level targeting | Targeting... | New Item | Security Group | ... | Locations... | group03.c24200.cit.lcl | [AD domain] | OK* in *Location* window | *Advanced... | Find Now*
10. Selected *[department name]_READ_DL | [department name]_MOD_DL* (held **Ctrl** for multiple selections)
11. Repeated step 10 for each [department name]
12. Clicked *OK* in *Select Group* window | *OK* in *Select Group* window | *OK* in *Targeting Editor* window | *OK* in *New Shared Printer Properties* window
13. Closed *Group Policy Management Editor* window
14. Right-clicked *Roaming Profiles | Enforced*

15. Repeated steps 1-14 for G03SRV01 and G03SRV02

Windows Activation

In order to activate Windows, the protocols were executed using a Microsoft activation key. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `cmd`
3. Right-clicked *Command Prompt*
4. Clicked *Run as Administrator*
5. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
6. Pressed **Enter**
7. Clicked *OK* in *Windows Script Host* window
8. Typed `slmgr /ipk NW6C2-QMPVW-D7KKK-3GKT6-VCFB2`
9. Pressed **Enter**
10. Clicked *OK* in *Windows Script Host* window
11. Typed `slmgr.vbs /ato`
12. Pressed **Enter**
13. Clicked *OK* in *Windows Script Host* window
14. Repeated steps 1-13 for G03SRV01 and G03SRV02

Latest Software Updates Verification

The protocols taken to ensure the domain controller was up to date with software updates. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Local Server*
4. Read *Last checked for updates* and *Last installed updates*
5. Repeated steps 1-4 for G03SRV01 and G03SRV02

Machines Domain Check

The protocols executed to verify the VM is a part of the Active Directory domain computers. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Local Server*
4. Verified *Domain* matched with *[AD domain]*
5. Clicked *Tools | Active Directory Users and Computers*
6. Expanded *[AD domain] | Computers*
7. Verified *[VM]*
8. Repeated steps 1-7 for G03SRV01 and G03SRV02

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APPENDIX A: PROBLEM SOLVING

The following Appendix section is dedicated to problems faced during the project completed in the microsoft windows administration project. Appendix A demonstrates proper system diagnostic procedures, troubleshooting, and research to ensure proper configuration of Microsoft Active Directory services. Each problem listed includes a description, possible solutions, solutions attempted, and the final solution. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

Problem 1: Incorrect OU creation and deletion

Problem Description: When creating OUs, groups, and users for the business scenario, some OUs and users were configured incorrectly. Instead of OUs being exclusive from each other, all the OUs were all created under a different parent OU, which complicated UNC paths and user locations. When attempting to delete the OUs, the prompt stated the OUs were protected from accidental deletion. The setting, which was not disabled when creating the OUs, did not allow for the correction of the original issue. When entering the properties page of the OU, there was no option to disable the *Protect object from accidental deletion* setting.

Problem Solutions: Ensuring the OUs were empty before deletion, ensuring there were no contents of the OU were protected from accidental deletion. Ensuring OU hierarchy was correct before creation of the OUs. Entering the Properties page of the OUs to delete the OUs. Ensuring Advanced Features was enabled under the View tab, then enter the properties page of the OU. Modify security settings to remove accidental delete protection.

Solutions Attempted: Ensuring OU hierarchy was correct before creation of the OUs.

Entering the Properties page of the OUs to delete the OUs. Ensuring Advanced Features was enabled under the View tab, then enter the properties page of the OU.

Final Solution: After Ensuring Advanced Features was enabled under the View tab of Active Directory Users and Computers, the Properties page of the OU was opened. After opening the Properties page, under the Object tab, *Protect object from accidental deletion* was unchecked, allowing for the OU to be deleted. Originally, when entering the Properties page of an OU, there was no Objects tab, therefore *Protect object from accidental deletion* could not be unchecked.

Problem 2: Block Regedit GPO Configuration

Problem Description: When configuring GPO permissions across each domain, Registry Edit was configured to be disabled through group policy. However, users logging in through the VM were still able to access Registry Edit after logon. The GPO was applied at the Domain level, however the GPO still did not block users from accessing Registry Edit.

Problem Solutions: Ensuring the *Disable Regedit* GPO was enforced, so there were no overriding/hierarchy issues. Confirming the GPO setting was correctly configured. Ensuring the under Security Filtering in the GPO, the intended users were added. Running the *gpresult /r* command to ensure the GPO applied to intended users.

Solutions Attempted: Ensuring the *Disable Regedit* GPO was enforced, so there were no overriding/hierarchy issues. Confirming the GPO setting was correctly configured. Running the *gpresult /r* command to ensure the GPO applied to intended users.

Final Solution: After confirming the GPO was applied to all intended users, the *Disable Regedit* GPO still did not function as intended. To solve the issue, Group Policy Management was opened. In Group Policy Management, the Group Policy Management Editor was opened for the *Disable Regedit* GPO. After navigating through user configuration, policies, administrative template, and system, “prevent access to registry editing tools” was right-clicked and edit was selected. Under the configurations page for the specific GPO, “Disabled” was originally selected, which was changed to “Enabled” to solve the issue of the *Disable Regedit* GPO not functioning as intended.

Problem 3: Incorrect Permissions for Folder Redirection

Problem Description: When configuring paths and GPOs for folder redirection, paths to the intended folder were not configured correctly and sharing permissions for the folder were not configured properly for all groups and users. Consequently, the misconfiguration meant signing in as a user on the VM, a folder including the user's desktop and documents were not created inside the intended folder for folder redirection. Although GPOs for folder redirection were properly configured, new folders were not created at user logon.

Problem Solutions: Ensuring sharing was enabled on the intended folder. Ensuring all intended groups for folder redirection were included group sharing. Ensuring sharing permission for added groups were correct. Ensuring all users intended for file redirection were included in their specific groups. Ensuring the UNC path led to the intended folder for folder redirection. Ensuring all folders and naming conventions were correct.

Solutions Attempted: Ensuring sharing was enabled on the intended folder. Ensuring the UNC path led to the intended folder for folder redirection. Ensuring the sharing permission for added groups were correct.

Final Solution: Ensuring the sharing permission for added groups were correct and sharing was enabled on the intended folder fixed the problem of folder redirection not working as intended. To solve the issue, the properties of the intended folder were opened. Next, under the sharing tab, advanced sharing was entered. In advanced sharing, the *share this folder* box was checked. Next, in the advanced sharing tab, permissions were opened. In the permissions tab, under *group or user names* the groups including all users were added. Finally, in the permissions tab under the

permissions for everyone section, the allow change box and allow read box were both checked.

APPENDIX B: USER CONFIGURATIONS

The following Appendix section is dedicated to demonstrating the User, Group, and VM configurations created during the microsoft windows administration project. Appendix B demonstrates how naming conventions, system settings, group scopes, and storage settings were all standardized across domains. The following steps were documented according to the computer G03SRV03; likewise, computers G03SRV01 and G03SRV02 were configured in the same fashion. Windows Server 2022 was expected to be booted before the first step was started.

User Configurations

First Name	Last Name	Logon Name	Password	Profile Path
Johnny	Utah	jutah	<Strong Password>	\\G03SRV01\Profiles\$

Figure 1: User Configuration Table displaying the general user conventions used in the Microsoft Windows Administration project.

Group Configurations

Group Name	Group Type	Group Scope
ACCG_Read_DL	Security Group	Domain Local
ACCG_Mod_DL	Security Group	Domain Local
ACCG_Manager_G	Security Group	Global

Figure 2: Group Configuration Table displaying the three general group types used in the Microsoft Windows Administration project.

VM Configurations

VM	VM OS	Dedicated	CPU Cores	Hard Drive	Disk Format
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Software		Memory		Size	
VMware Workstation	Windows 11 Education	4 GB	2 Cores	64 GB	Virtual disk split into multiple files

Figure 3: VM hardware configuration table displaying the dedicated hardware allotted to the VM.

VM Storage Location	Network Mode	Domain Membership	VM Computer Name	Activation	Software Updates
D:Drive	Bridge	One VM per Domain	G03SxWIN11	Purdue KMS	Latest Windows

Figure 4: VM software configuration table displaying the conventions and allocations of the Windows 11 VM.